

User Story	Acceptance Criteria	Feature
As a user, I want the system to automatically receive and log new credit card transactions.	The app receives transactions in real time; Each transaction is timestamped and stored.	Transaction Receiver
As a user, I want invalid or incomplete transactions to be rejected before analysis.	Transactions missing fields like amount or merchant ID are dropped; Dropped transactions are logged with a reason	Transaction Sanitizer
As a user, I want all transactions to be formatted consistently before being processed by the model.	Currency, timestamp, and amount fields are standardized; Cleaned transactions are passed to the model.	Transaction Sanitizer
As a user, I want each transaction to be scored for fraud likelihood.	Model returns a fraud probability score between 0 and 1; Transactions with missing features are skipped and logged.	Fraud Detection Model Wrapper
As a user, I want the app to classify transactions as fraudulent or safe based on a configurable threshold.	The threshold value can be set manually; Transactions above threshold are marked as 'Fraudulent'.	Fraud Detection Model Wrapper
As a user, I want to be notified immediately when a fraudulent transaction is detected	Notification includes merchant, amount, and time; Notification is delivered within 5 seconds of detection.	Notification Service
As a user, I want legitimate transactions to be logged for auditing purposes.	Non-fraudulent transactions are recorded with timestamp and merchant; No notification is sent.	Transaction Receiver
As a user, I want duplicate transactions to be ignored.	Duplicate detection based on transaction ID; Only first instance is processed; Duplicates logged.	Transaction Sanitizer
As a user, I want to mark a transaction as 'Not Fraudulent' if it was wrongly flagged.	User can flag false positives through the app; Feedback is stored for model retraining.	Notification Service
As a data scientist, I want to export flagged transactions for model retraining.	Fraudulent and legitimate labels are stored; Export available as JSON or CSV.	Fraud Detection Model Wrapper
As a user, I want to choose how to receive alerts (SMS, push, or email).	User selects preferred channel; Notifications are routed correctly.	Notification Service
As a user, I want to view a list of past transactions and their fraud status.	Transactions can be filtered by date or status; History includes fraud score and decision.	Notification Service
As a developer, I want a way to check if all system components are running.	A health endpoint returns component statuses; Returns 200 OK only when all services are operational.	Transaction Receiver
As a developer, I want to track latency between transaction receipt and notification.	Start and end timestamps recorded for every transaction; Average latency reported in logs.	Transaction Receiver
As a user, I want to see a summary report of detected fraud over time.	Report shows counts by day/week/month; Fraud rate percentage is calculated.	Notification Service